3. Candidate 1 — rat prefrontal-cortex layer V pyramidal neuron

This is probably my favourite after doing this search

There is an unusually clean experimental paper specifically studying **layer V pyramidal neurons in rat medial/prefrontal cortex** and pharmacologically modifying their M-current.

The researchers recorded directly from these cells using whole-cell patch clamp.

They used:

- **XE991** — Kv7/M-channel blocker;
- **retigabine** — M-channel opener;
- **QO58L** — another M-channel opener.

They measured both the M-current itself and neuronal firing. XE991 decreased the M-current and increased spike frequency, while retigabine/QO58L increased M-current and reduced firing.

PubMed Central (...

Even better, the paper has exactly the sort of figures your project wants:

Figure 1

- image of the specific layer V pyramidal cell;
- voltage-clamp protocol;
- control vs XE991 M-current traces;
- control vs retigabine current traces.

Figure 2

- actual action-potential traces;
- control vs blocker;
- control vs opener;
- firing frequency versus injected current;
- interspike intervals.

PubMed Central (...

That's extraordinarily convenient.

4. What would the ODE model look like?

You don't need a morphologically realistic neuron with hundreds of compartments.

There are deliberately **minimal single-compartment Hodgkin–Huxley models of regular-spiking cortical pyramidal neurons** containing

$$I_{\mathrm{Na}}, \quad I_{\mathrm{Kd}}, \quad I_M, \quad I_L.$$

Pospischil et al. developed minimal HH-type models fitted to experimental cortical neuron behaviour, including regular-spiking pyramidal neurons. Their regular-spiking model uses I_M specifically to reproduce spike-frequency adaptation. PubMed +1

A corresponding system is approximately

$$C_m \dot{V} = I_{\mathrm{app}} - g_L(V - E_L) - \bar{g}_{\mathrm{Na}} m^3 h (V - E_{\mathrm{Na}}) - \bar{g}_K n^4 (V - E_K) - \bar{g}_M p (V - E_K),$$

with

$$\dot{m} = f_m(V, m),$$
$$\dot{h} = f_h(V, h),$$
$$\dot{n} = f_n(V, n),$$
$$\dot{p} = f_p(V, p).$$

So that's a **5D deterministic ODE**

$$\boxed{(V, m, h, n, p)}$$

which is exactly at the upper end of your guideline's preferred $3 - 5D$ range. Project Pitch - Guidelines

Or because sodium activation m is extremely fast, use

$$m = m_\infty(V)$$

to obtain

$$\boxed{(V, h, n, p)}$$

as a **4D model**.

I actually like that even more for continuation.

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5. And here's where your group could add its own spin

Rather than saying:

> XE991 decreases g_M.

you could actually investigate **how much M-current suppression is required to change neuronal dynamics.**

Write

$$I_M = g_M^{\text{eff}} p(V - E_K)$$

with

$$g_M^{\text{eff}} = (1 - \rho) g_M, \qquad 0 \le \rho \le 1,$$

where

↓

$$\rho = 0$$

means no block and

$$\rho = 1$$

means complete block.

Then use PyDSTool to continue in

$$(I_{\text{app}}, \rho).$$

You'd get something conceptually like:

resting	tonic spiking	burst/hyperexcitable

in the parameter plane.

And you'd be able to answer a biological question:

> **How does pharmacological suppression of the Kv7/M-current shift the threshold separating quiescent and repetitively firing behaviour in layer V PFC pyramidal neurons?**

That's a strong MATH3888 question.

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6. You could make retigabine even more interesting mathematically

This is one thing I particularly like about choosing this cell.

An M-channel opener such as retigabine doesn't simply have to be modelled as

$$g_M \uparrow .$$

Experimentally, retigabine modifies the voltage dependence of M-channel activation, including shifting activation in the hyperpolarising direction. PubMed Central (... +1

Suppose

$$p_\infty(V) = \frac{1}{1 + \exp(-(V - V_{1/2})/k)}.$$

You could represent the drug by

$$V_{1/2} \longrightarrow V_{1/2} - \Delta.$$

Now compare two physiological manipulations:

$$\boxed{\text{XE991: reduce } g_M}$$

versus

$$\boxed{\text{retigabine: shift M-channel activation}}$$

and ask whether they move the bifurcation boundary symmetrically or in qualitatively different ways.

That is a much more interesting mathematical project than merely “multiply a conductance by 0.5.”

7. The one caveat with this option

The experimental study is **specifically rat PFC layer V pyramidal neurons**, whereas the minimal Pospischil-type model is a more generic regular-spiking cortical pyramidal model fitted across cortical preparations, not a bespoke model of those exact PFC layer V cells. PubMed

I wouldn't hide that.

Instead, you could actually make it part of the project:

$$\boxed{\text{adapt/calibrate a minimal cortical HH model to PFC layer V data}}$$

For example tune a small number of parameters so that the baseline model approximately matches experimental:

$$V_{\text{rest}}, \qquad \text{rheobase}, \qquad F-I \text{ curve}, \qquad \text{adaptation}.$$

Then **don't overfit** dozens of parameters.

That is a legitimate modelling contribution and arguably gives you more ownership over the model.